Syllabus Details

Syllabus ID:	13156
Syllabus Name:	**Game Programming with C#_Lập trình Game với C#**
Course Name English:	**Game Programming with C#**
Subject Code:	**PRU213**
Learning-Teaching Method:	Social Constructivism,In-class lecture,Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 104h self-study
Pre-Requisite:	PRO192
Description:	This comprehensive Basic Unity Training Program is designed to equip participants with essential skills in Unity game development. Starting from the fundamentals of Unity interface and scripting in C#, the course progresses to cover advanced topics such as physics, animations, and user interface design. Through a combination of lectures, hands-on labs, and real-world projects, participants will gain practical experience and confidence in using Unity for game development.
StudentTasks:	- Students must attend at least 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises, assignments and labs given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the https://flm.fpt.edu.vn/ for up-to-date course information
Tools:	- Internet - Visual Studio 2019 or later (C# programming language) - Unity software (latest version) - Students use Unity with their personal email for educational purposes only
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	932/QĐ-ĐHFPT dated 08/22/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	8/22/2025

3 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Online Tutorials: https://www.youtube.com/user/Unity3D						False	False	True	
2	Unity Learn Platform:https://learn.unity.com						True	False	True	
3	Unity Official Documentation: https://docs.unity3d.com/Manual/index.html						False	False	True	

7 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand the Unity interface and basic tools for game development
2	CLO2	Write and implement C# scripts for game logic and interactions.
3	CLO3	Create and manipulate game objects, animations, and physics components.
4	CLO4	Design user interfaces and handle user input effectively.
5	CLO5	Develop small games and interactive applications using Unity.
6	CLO6	Troubleshoot and debug common Unity issues.
7	CLO7	Collaborate effectively in a team environment for game development projects.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	Course Introduction 1.1 Introduction to Unity Interface and Navigation	Offline	CLO1	T	Slide of Introduction to Unity Interface and Navigation	PRU213	Read the slide of Introduction to Unity Interface and Navigation	
2	Course Introduction 1.1 Introduction to Unity Interface and Navigation	Offline	CLO1	T	Slide of Introduction to Unity Interface and Navigation	PRU213	Read the slide of Introduction to Unity Interface and Navigation	
3	1.2 Basic Game Object Manipulation and Transformations	Offline	CLO1	T	Slide of Basic Game Object Manipulation and Transformations		Read the slide of Basic Game Object Manipulation and Transformations	
4	1.2 Basic Game Object Manipulation and Transformations	Offline	CLO1	T	Slide of Basic Game Object Manipulation and Transformations		Read the slide of Basic Game Object Manipulation and Transformations	
5	1.3 Understanding Unity's Component System	Offline	CLO1	T	Slide of Understanding Unity's Component System		Read the slide of Understanding Unity's Component System	
6	1.3 Understanding Unity's Component System	Offline	CLO1	T	Slide of Understanding Unity's Component System		Read the slide of Understanding Unity's Component System	
7	1.4 Scene Creation and Management	Offline	CLO1	T	Slide of Scene Creation and Management		Read the slide of Scene Creation and Management	
8	1.4 Scene Creation and Management	Offline	CLO1	T	Slide of Scene Creation and Management		Read the slide of Scene Creation and Management	
9	1.5 Introduction to Physics and Colliders	Offline	CLO1	T	Slide of Introduction to Physics and Colliders		Read the slide of Introduction to Physics and Colliders	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
10	1.5 Introduction to Physics and Colliders	Offline	CLO1	T	Slide of Introduction to Physics and Colliders		Read the slide of Introduction to Physics and Colliders	
11	1.5 Introduction to Physics and Colliders	Offline	CLO1	T	Slide of Introduction to Physics and Colliders		Read the slide of Introduction to Physics and Colliders	
12	1.5 Introduction to Physics and Colliders	Offline	CLO1	T	Slide of Introduction to Physics and Colliders		Read the slide of Introduction to Physics and Colliders	
13	Quiz 1	Offline	CLO1	U	- Content module 1		- Review module 1 '- Do quiz 1	
14	Quiz 1	Offline	CLO1	U	- Content module 1		- Review module 1 '- Do quiz 1	
15	2.1 Introduction to C# Programming in Unity Introduction to Final Project	Offline	CLO2	T	Slide of Introduction to C# Programming in Unity Introduction to Final Project		Read the slide of Introduction to C# Programming in Unity Introduction to Final Project	
16	2.1 Introduction to C# Programming in Unity Introduction to Final Project	Offline	CLO2	T	Slide of Introduction to C# Programming in Unity Introduction to Final Project		Read the slide of Introduction to C# Programming in Unity Introduction to Final Project	
17	2.2 Variables, Data Types, and Operators	Offline	CLO2	T	Slide of Variables, Data Types, and Operators		Read the slide of Variables, Data Types, and Operators	
18	2.2 Variables, Data Types, and Operators	Offline	CLO2	T	Slide of Variables, Data Types, and Operators		Read the slide of Variables, Data Types, and Operators	
19	2.3 Control Structures: Loops and Conditional Statements	Offline	CLO2	T	Slide of Control Structures: Loops and Conditional Statements		Read the slide of Control Structures: Loops and Conditional Statements	
20	2.3 Control Structures: Loops and Conditional Statements	Offline	CLO2	T	Slide of Control Structures: Loops and Conditional Statements		Read the slide of Control Structures: Loops and Conditional Statements	
21	2.3 Control Structures: Loops and Conditional Statements	Offline	CLO2	T	Slide of Control Structures: Loops and Conditional Statements		Read the slide of Control Structures: Loops and Conditional Statements	
22	2.4 Functions and Methods	Offline	CLO2	T	Slide of Functions and Methods		Read the slide of Functions and Methods	
23	2.4 Functions and Methods	Offline	CLO2	T	Slide of Functions and Methods		Read the slide of Functions and Methods	
24	2.5 Scripting Interactivity: Mouse and Keyboard Input	Offline	CLO2	T	Slide of Scripting Interactivity: Mouse and Keyboard Input		Read the slide of Scripting Interactivity: Mouse and Keyboard Input	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
25	2.5 Scripting Interactivity: Mouse and Keyboard Input	Offline	CLO2	T	Slide of Scripting Interactivity: Mouse and Keyboard Input		Read the slide of Scripting Interactivity: Mouse and Keyboard Input	
26	Quiz 2 and Lab 1	Offline	CLO2, CLO5	U	- Content module 2		- Review module 2 '- Do quiz 2 - Present lab 1	
27	Quiz 2 and Lab 1	Offline	CLO2, CLO5	U	- Content module 2		- Review module 2 '- Do quiz 2 - Present lab 1	
28	3.1 Introduction to Unity Animation System	Offline	CLO3	T	Slide of Introduction to Unity Animation System		Read the slide of Introduction to Unity Animation System	
29	3.1 Introduction to Unity Animation System	Offline	CLO3	T	Slide of Introduction to Unity Animation System		Read the slide of Introduction to Unity Animation System	
30	3.2 Animation Scripting and Interactions	Offline	CLO3	T	Slide of Animation Scripting and Interactions		Read the slide of Animation Scripting and Interactions	
31	3.2 Animation Scripting and Interactions	Offline	CLO3	T	Slide of Animation Scripting and Interactions		Read the slide of Animation Scripting and Interactions	
32	3.3 User Interface Design: Buttons, Text, and Images	Offline	CLO3	T	Slide of User Interface Design: Buttons, Text, and Images		Read the slide of User Interface Design: Buttons, Text, and Images	
33	3.3 User Interface Design: Buttons, Text, and Images	Offline	CLO3	T	Slide of User Interface Design: Buttons, Text, and Images		Read the slide of User Interface Design: Buttons, Text, and Images	
34	3.4 User Input Handling: UI Events and Touch Input	Offline	CLO3	T	Slide of User Input Handling: UI Events and Touch Input		Read the slide of User Input Handling: UI Events and Touch Input	
35	3.4 User Input Handling: UI Events and Touch Input	Offline	CLO3	T	Slide of User Input Handling: UI Events and Touch Input		Read the slide of User Input Handling: UI Events and Touch Input	
36	3.5 Designing Interactive UI Systems	Offline	CLO3	T	Slide of Designing Interactive UI Systems		Read the slide of Designing Interactive UI Systems	
37	3.5 Designing Interactive UI Systems	Offline	CLO3	T	Slide of Designing Interactive UI Systems		Read the slide of Designing Interactive UI Systems	
38	3.5 Designing Interactive UI Systems	Offline	CLO3	T	Slide of Designing Interactive UI Systems		Read the slide of Designing Interactive UI Systems	
39	Quiz 3	Offline	CLO3	U	- Content module 3		- Review module 3 '- Do quiz 3	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
40	4.1 Unity's Physics Engine: Rigid Bodies and Forces	Offline	CLO4	T	Slide of Unity's Physics Engine: Rigid Bodies and Forces		Read the slide of Unity's Physics Engine: Rigid Bodies and Forces	
41	4.1 Unity's Physics Engine: Rigid Bodies and Forces	Offline	CLO4	T	Slide of Unity's Physics Engine: Rigid Bodies and Forces		Read the slide of Unity's Physics Engine: Rigid Bodies and Forces	
42	4.2 Implementing Interactive Physics-Based Puzzles	Offline	CLO4	T	Slide of Implementing Interactive Physics-Based Puzzles		Read the slide of Implementing Interactive Physics-Based Puzzles	
43	4.2 Implementing Interactive Physics-Based Puzzles	Offline	CLO4	T	Slide of Implementing Interactive Physics-Based Puzzles		Read the slide of Implementing Interactive Physics-Based Puzzles	
44	4.3 Handling Collisions and Triggers	Offline	CLO4	T	Slide of Handling Collisions and Triggers		Read the slide of Handling Collisions and Triggers	
45	4.3 Handling Collisions and Triggers	Offline	CLO4	T	Slide of Handling Collisions and Triggers		Read the slide of Handling Collisions and Triggers	
46	4.4 Game Logic and Scoring Systems	Offline	CLO4	T	Slide of Game Logic and Scoring Systems		Read the slide of Game Logic and Scoring Systems	
47	4.4 Game Logic and Scoring Systems	Offline	CLO4	T	Slide of Game Logic and Scoring Systems		Read the slide of Game Logic and Scoring Systems	
48	4.5 Scripting Animations and Physics Interactions	Offline	CLO4	T	Slide of Scripting Animations and Physics Interactions		Read the slide of Scripting Animations and Physics Interactions	
49	4.5 Scripting Animations and Physics Interactions	Offline	CLO4	T	Slide of Scripting Animations and Physics Interactions		Read the slide of Scripting Animations and Physics Interactions	
50	Quiz 4 and Lab 2	Offline	CLO4, CLO5	U	- Content module 4		- Review module 4 '- Do quiz 4 - Present lab 2	
51	Quiz 4 and Lab 2	Offline	CLO4, CLO5	U	- Content module 4		- Review module 4 '- Do quiz 4 - Present lab 2	
52	5.1 Applying Scripting and Physics to Game Projects	Offline	LO6	T	Slide of Applying Scripting and Physics to Game Projects		Read the slide of Applying Scripting and Physics to Game Projects	
53	5.2 Implementing Animations and UI in Projects	Offline	LO6	T	Slide of Implementing Animations and UI in Projects		Read the slide of Implementing Animations and UI in Projects	
54	5.2 Implementing Animations and UI in Projects	Offline	LO6	T	Slide of Implementing Animations and UI in Projects		Read the slide of Implementing Animations and UI in Projects	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
55	5.3 Debugging and Troubleshooting in Unity	Offline	LO6	T	Slide of Debugging and Troubleshooting in Unity		Read the slide of Debugging and Troubleshooting in Unity	
56	Final Project Presentations and Feedback	Offline	LO7	U	Student's projects and slides		Present projects	
57	Final Project Presentations and Feedback	Offline	LO7	U	Student's projects and slides		Present projects	
58	Final Project Presentations and Feedback	Offline	LO7	U	Student's projects and slides		Present projects	
59	Final Project Presentations and Feedback	Offline	LO7	U	Student's projects and slides		Present projects	
60	Review	Offline	All CLOs	U	All Slides		Review studied materials	

0 Constructive question(s)

4 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Final project	on-going	1	30.0%	>0	6-8 weeks	CLO7	theory present or project		technical aspect of studied topics	- Guided by instructor - Application or Theory - Work in group	
2	Labs	on-going	2	20.0%	>0	1 - 2 weeks	CLO5	theory present or project		technical aspect of studied topics	- Guided by instructor - Application or Theory - Work in group	
3	Quizzes	on-going	4	20.0%	>0	30'	CLO1, CLO2, CLO3, CLO4	Multiple choicesMarked by Computer	20	concepts, algorithms; all studied chapters	in class, by instructor	

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
4	Final exam	Final exam	1	30.0%	4	60'	All CLOs	Multiple choices Marked by Computer	50	concepts, algorithms; all studied chapters	by exam board, using computer	The exam questions must be updated or different at least 70% to the previous ones.